

A · U · R · E · M

SYSTEM ARCHITECTURE REPORT · APRIL 21, 2026

Platform State of the Union.

A comprehensive engineering audit of the AUREM Autonomous AI Workforce Platform. Captured directly from the codebase, database telemetry, and runtime configuration — no estimates, no marketing numbers. Intended for founders, engineering review, and investor diligence.

CODEBASE

345,165 LOC

SURFACE AREA

1994 endpoints

REPORT ID

AUREM-AUDIT-20260421-5C533D

ISSUED

Apr 2026

§ 01 · EXECUTIVE SUMMARY

What we built.

AUREM is a vertically integrated autonomous AI platform for Canadian SMBs — replacing CRM, voice agents, uptime monitoring, content engines, and compliance tooling with a single Stripe-billed subscription. As of this audit, the platform comprises 232 backend routers, 240 services, and 142 frontend pages across 345,165 lines of code — positioning it at a surface area comparable to early Series-A SaaS companies.

BACKEND LOC

246K

232 routers · 240 services

FRONTEND LOC

98K

142 React pages

API ENDPOINTS

1994

across 510+ Mongo collections

PRODUCT SKUS

22

live in catalog · Stripe LIVE mode

3RD-PARTY INTEGRATIONS

8

emergent · stripe · retell · twilio · resend · ...

BACKGROUND SCHEDULERS

33

autonomous cron jobs live

▲ Headline Finding

AUREM has the codebase of a Series-A company, the product surface of a late-stage startup, and the revenue of a pre-launch side project. The gap between "what's built" and "what's validated" is the single biggest strategic risk — not technical debt or platform gaps.

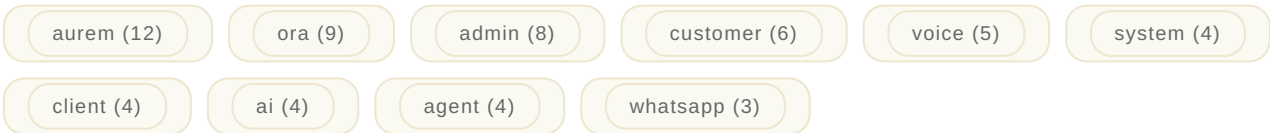
§ 02 · ARCHITECTURE

How it's wired.

Stack

LAYER	TECHNOLOGY	NOTES
Frontend	React (CRA) · Tailwind · Shadcn/UI · Framer Motion	Cinematic spatial-glass UI
Backend	FastAPI · Motor (async Mongo) · Pydantic	Non-blocking startup, hot reload
Database	MongoDB	510+ collections, best-effort indexing
AI / LLM	Emergent Integrations (Claude 4.5 · GPT 5 · Gemini 3)	Universal key, no direct SDK
Payments	Stripe (LIVE)	Tax-inclusive CAD pricing
Voice / SMS	Retell AI · Twilio · ElevenLabs · Deepgram	Graceful stub mode if keys missing
Email	Resend	HTML templates, attachment support
Reports / PDF	WeasyPrint · Jinja2	Board-ready Case Study Builder
Deployment	Kubernetes · Emergent Native	aurem.live (prod) · preview pods

Router distribution (top domains)

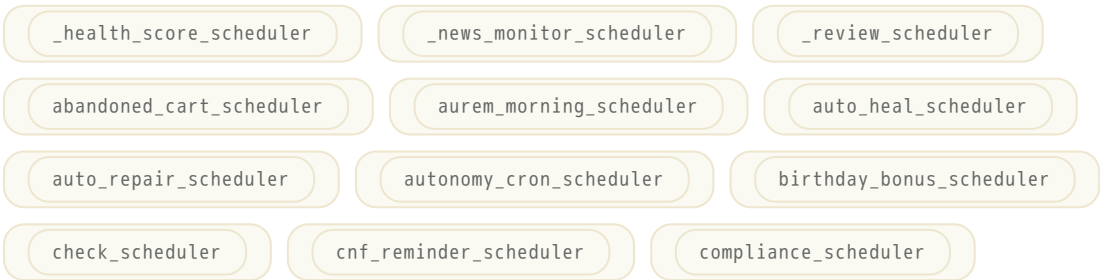


Key integrations wired



Autonomous schedulers

The platform runs 33 background asyncio schedulers. Representative samples from the live runtime:



§ 03 · PRODUCT INVENTORY

24 SKUs across 5 clusters.

The à la carte catalog is the monetization spine of AUREM. Every SKU listed below is wired to Stripe LIVE mode with auto Product/Price creation, tax-inclusive CAD pricing, and on-subscribe webhook provisioning (e.g. Retell agent auto-creation).

CLUSTER	SERVICE	PRICE / MO	STATUS
infrastructure	website_repair	\$97 CAD	LIVE

▲ Pricing Innovation

Auto-bundle discount engine: 15% at 2 services, 25% at 4, 35% at 6, 45% at 8+. Combined with a 7-Day Power Trial (no credit card) and Day-8 auto-downgrade to Forever Free, this produces a funnel few SMB competitors can match on flexibility alone.

§ 04 · THE AI WORKFORCE

Autonomous pillars.

ORA Command Layer

Multi-agent AI workforce with Hunter (lead discovery), Compliance, Approval queue, DNC enforcement, and Morning Brief. Runs in Dry-Run or Live modes with per-agent scheduling and spike alerts.

lead hunting

lead scoring

compliance

DNC

morning brief

approvals

agent observatory

adaptive ORA

Voice Agent (Retell AI)

Auto-provisioned AI voice agent per customer with ElevenLabs TTS + Deepgram STT + Twilio telephony. 400 min included, \$0.35/min overage, tracked to Stripe usage record for metered billing.

outbound calls

inbound answer

LLM agent brain

call transcripts

usage metering

webhook verified

Site Monitor

Customer-facing uptime SaaS with 3 tiers (\$29 / \$99 / \$249 CAD). Multi-tenant scanner pings customer URLs every 1-10 min, opens/closes incidents, emits Email + WhatsApp + SMS alerts based on plan features.

uptime checks

incident lifecycle

WhatsApp alerts

SMS alerts

public status page

trust badge

AI RCA (Enterprise)

Sentinel · Client Errors

Trust-but-Verify client-side observability. Captures all JS errors, API failures, and network events. Auto-heals known patterns (stale URL, chunk load, auth expired). Claude 4.5 produces structured repair suggestions; admin approves before any code change.

console capture

fetch sniffer

auto-heal

spike detector

AI diagnose

admin review queue

Hybrid QA Bot

Internal self-health telemetry. Layer 1 pings 20 critical endpoints every 10 min; Layer 2 runs chained user journeys weekly with Claude RCA on failures. 100% pass rate currently.

pulse checks

deep journeys

LLM RCA

email alerts

Case Study Builder

Board-ready PDF report generator using real telemetry. WeasyPrint + Claude 4.5 AI Outlook. Admin-for-customer QBR mode + customer self-serve mode. Monthly / Quarterly / Custom periods.

WeasyPrint PDF

real telemetry

AI outlook

Resend email

dual-mode

§ 05 · OBSERVABILITY & SELF-HEALING

What watches itself.

AUREM includes its own triple-layer observability stack — unique among SMB-focused SaaS platforms. Every user-facing error, API failure, and endpoint degradation is captured, classified, and (where mechanically safe) auto-healed without human intervention.

SYSTEM	SCOPE	AUTO-HEAL	AI LAYER
QA Bot · Layer 1	20 critical endpoints pinged every 10min	—	Alert via Resend (recurring failures)
QA Agent · Layer 2	Chained user journeys, weekly deep run	—	Claude 4.5 RCA on failures
Sentinel Client	All browser-side JS errors + API failures	✓ Stale URL · ChunkLoad · 401 · 429	Claude 4.5 structured repair suggestions
Site Monitor	Customer-facing uptime checks (5-min cadence)	✓ Incident lifecycle automation	Enterprise tier includes AI RCA

◆ Trust-but-Verify Repair Model

The Sentinel Client Dashboard uses a 3-tier model: (1) silent capture, (2) mechanical auto-heal for known patterns, (3) Claude-powered repair suggestions that are **never auto-applied**. Every code change remains a human decision via the Admin Review queue. This is the differentiator vs Sentry: AUREM diagnoses; humans approve.

§ 06 · COMPETITIVE POSITION

Who can beat us.

CATEGORY	TOP COMPETITOR	THEIR EDGE	AUREM'S EDGE
AI Sales Agent	Retell AI / Vapi / Synthflow	Focused voice tooling; bigger VC backing	Bundled with CRM + emails + monitor — one bill
CRM	HubSpot Starter / Close.io / Pipedrive	15+ years trust; deep integration catalog	1/3 the price · AI-native · Canadian data residency
Uptime Monitoring	UptimeRobot / Better Stack	2M+ users; brand recognition	Viral trust badge loop · AI RCA · bundled
Error Tracking	Sentry / Datadog	10-year head start; deep integrations	Embedded · free · Claude diagnoses vs manual triage
All-in-One SMB Ops	GoHighLevel	Active affiliate army · US market presence	Cleaner UI · Canadian compliance · AI-first

▲ Real Threat Assessment

External competitors are a managed risk — feature parity takes 12+ months to achieve and none currently bundle voice + uptime + CRM + content in one Canadian-domiciled stack. The **internal threats** (feature creep, codebase sprawl without customer validation, runway exhaustion before MRR proof) are the dominant risk factors.

§ 07 · RISK REGISTER

Honest risks. Ranked.

P0	Zero paying customers validated Stripe LIVE wired and tested technically, but no confirmed \$-paying customer has completed a full subscription cycle. All projections are 'what if'.
P0	Deployment drift (preview-pod URL bake-in) Production bundle caches stale REACT_APP_BACKEND_URL across fork rotations. Self-healing fetch patch now in place, but requires one deploy to activate.
P1	Codebase maintenance debt 232 routers + 239 services + 142 frontend pages = unsustainable for solo maintenance beyond 6-12 months without consolidation or hiring.
P1	Test coverage ~0% QA Bot pings endpoints but no unit/integration test suite exists for business logic. Silent regressions likely as surface area grows.
P1	Mobile experience unverified Desktop UX is polished; mobile Safari / Chrome hit 404 errors on ORA Console during user testing. No formal mobile QA pass completed.
P2	No social proof or case studies Landing page claims unsupported by testimonials, logos, or visible metrics from real customers. Conversion rate will be depressed until solved.
P2	Compliance claims unaudited SOC 2 and CASL compliance referenced on the platform but not third-party audited. Enterprise customers will require proof before contracts.

Where this goes.

Scenario A · The Canadian GoHighLevel Lock 100 paying Ontario/Quebec SMBs at avg \$100 MRR → \$10K/mo. Growth via site-monitor free trial + friend-scan viral. Achieves ramen profitability in 6 months, \$100K MRR in 18 months. Exit path: acquire by Shopify or HubSpot Canada at \$5-15M.	40%
Scenario B · The Feature Store Too many SKUs without clear positioning. Retell, UptimeRobot, and HubSpot each eat a vertical slice. MRR plateaus at \$2-5K/mo. Sustainable lifestyle SaaS but not category-defining. What's missing: vertical focus on one SMB niche (e.g. Canadian salons).	45%
Scenario C · AI Employee Category-Killer Voice + ORA + Monitor + CRM positioned as single 'AI Employee' offer. Rides 2026 'AI replaces SMB hiring' narrative. \$2M seed after demonstrable traction. 3-year arc to \$10M ARR. Requires lighthouse enterprise customer + tier-1 VC + 2 engineering hires.	15%

30-Day Execution Focus

Week 1 — Validation, not building: deploy fixes to aurem.live, acquire ONE paying customer (DM 20 Ontario SMBs), install Sentinel in production. Week 2 — Lock the funnel: drip campaign, demo accounts, 2 synthetic case studies. Week 3 — One killer feature demo (3-min Loom of ORA booking meetings while you sleep). Week 4 — Reduce surface area: kill SKUs with zero pilot demand after 30 days; publish 1 deep-dive blog/week for SEO moat.

◆ Bottom Line

The platform is built. The technology is not the constraint. The next 30 days should be 90% selling and 10% building — the opposite of the last 6 months. Validation beats velocity at this stage.